5 OCTOBER 1982
SUPERSEDING
MIL-S-4510D (USAF)
31 DECEMBER 1969

MILITARY SPECIFICATION

SPOTLIGHT, INCANDESCENT, ROOF-MOUNTED, AUTOMOTIVE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers roof-mounted spotlights of 6, 12, and 24 volt ratings, using PAR 46, screw-terminal lamps of the proper voltage and power for use on commercial vehicles.

1.2 Classification. The spotlights covered in this specification shall be of the following types, classes, and sizes as specified (see 6.2):

Type I - To be used after mounting with no further treatment.

Type II - Primed to receive painting to match vehicle paint.

Class A - 6" control arm.

Class B - 23" control arm.

*1.3 Part number. Specification part number for items described in this specification will be formulated as shown in section 6.3.

2. APPLICABLE DOCUMENTS

*2.1 Government documents.

*2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center/MMEDO, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Federal

	QQ-P-416	Plating, Cadmium (Electrodeposited)
	QQ-S-571	Solder, Tin Alloy, Lead-Tin Alloy and Lead Alloy
	PPP-B-566	Box, Folding, Paperboard
	PPP-B-585	Box, Wood, Wire Bound
*	PPP-T-60	Tape, Packaging, Waterproof
	PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated
	PPP-B-601	Boxes, Wood, Cleated-Plywood
	PPP-B-621	Box, Wood, Nailed and Lock-Corner
*	PPP-B-636	Boxes, Shipping, Fiberboard
*	PPP-B-665	Boxes, Paperboard, Metal Edged and Components
	PPP-B-676	Boxes, Set-up

Military

	MIL-P-116	Preservation, Methods of
*	MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloy
	MIL-S-6872	Soldering Process, General Specification for
	MIL-A-8625	Anodic Coating for Aluminum and Aluminum Alloys

STANDARDS

Military

	MIL-STD-129	Marking for Shipment and Storage
	MIL-STD-130	Identification Marking of U.S. Military Property
*	MIL-STD-808	Finish, Protective and Codes for Finishing Schemes for Ground and Ground Support Equipment
*	MIL-STD-889	Dissimilar Metals
*	MS18005	Lamp, Incandescent and Sealed Beam, PAR 46 Spot Lamp, 2 Screw Terminal Type
*	MS24517	Lamp, Incandescent, PAR 46 bulb, Screw Terminal Base, Taxing
*	MS25241	Lamp, Incandescent PAR 46 Bulb, Screw Terminal Base Landing

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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SAE J575 Tests for Motor Vehicle Lighting Devices & Components

(Copies of publications may be obtained from the Society of Automotive Engineers, Inc., 2 West 39th St., New York, New York 10018.)

Official Classification Committee

Uniform Freight Classification Rules

(Application for copies of the above publication should be addressed to the Official Classification Committee, 1 Park Avenue, New York, New York 10016.)

American Trucking Association

National Motor Freight Classification Rules and Container Specifications

(Application for copies of the above publication should be addressed to the American Trucking Association, 1424 16th St. N.W. Washington D.C.)

3. REQUIREMENTS.

3.1 Preproduction. Preproduction samples shall be furnished for inspection and testing as specified in 4.3.

3.2 Materials. Materials shall be the suppliers standard commercial material with such changes as are necessary to meet the requirements of this specification.

3.2.1 Solder. Solder employed in making electrical connections shall be a rosin core type conforming to compositions SN50, SN60, or SN70 of QQ-S-571.

3.2.1.1 Soldering. Soldering shall meet the requirements of MIL-S-6872.

*3.2.2 Dissimilar metals. Except where necessary to complete an electrical circuit contact between dissimilar metals shall be avoided. Where such contact is unavoidable, metals shall be treated to prevent deterioration by chemical or electrochemical reaction with its environment. Requirements for protecting coupled dissimilar metals shall be in accordance with MIL-STD-889.

3.3 Design and construction. The spotlight shall be the manufacturers standard item, modified as required to meet the requirements of this specification.

3.3.1 Movement. The movement of the lamp housing shall be controlled by one hand operation of the control handle (knobs). The lamp housing shall move in a zimuth thru an arc of not less than 270 degrees and shall move in elevation from vertical to not less than 35° below horizontal.

3.3.2 Bulb. The spotlight shall be designed for insertion of bulbs of the PAR 46, screw terminal configuration. The bulb specified (see Table I and 6.2) shall determine the size of the spotlight.

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3.4 Performance. The spotlight shall be capable of operating satisfactorily under the following conditions:

a. Temperature ranging from minus 54 degrees centigrade to plus 71 degrees centigrade.

b. Exposure to salt-sea atmosphere.

c. Vibration incident to service use.

d. Fungus growth as encountered in humid climates.

e. Rainfall as encountered in any locale.

f. Sand and dust conditions as encountered in arid locales.

3.5 Components. The manufacturer's standard commercial components shall be used with such modifications as are necessary to meet the dimensional and performance requirements of this specification. The spotlight shall consist of the following major components:

a. Lamp housing (see 3.5.1).

b. Lamp (see 3.5.2).

c. Support Arm (see 3.5.3).

d. Control arm and handle (see 1.2, 6.2 and 3.5.4).

e. Wiring (see 3.5.5).

* f. Roof seal (see 3.5.6).

3.5.1 Lamp housing. The lamp housing shall be fabricated of metal and shall be designed to hold and position PAR 46 screw terminal lamps. The indexing notch for the lamp shall be positioned to locate the widest divergence of the beam in the horizontal plane. The maximum dimension for the housing shall be 6 5/16" diameter and 6" length.

*3.5.2 The lamp. The lamp shall be a PAR 46, screw terminal base specified in Table I.

TABLE I. Size, sealed beam trade numbers and operating characteristics.

SIZE	TRADE NO.	DESIGN VOLTS	WATTS	NOMINAL C.P.	BEAM HORIZ	SPREAD VERT	RATED LIFE* HOURS	MILITARY STANDARDS
4535	4535	6.4	30	95,000	5-1/2°	4°	100	MS18005-4535
4533	4533	6.5	40	50,000	7°	7°	50	MS18005-4533
4435	4435	12.8	30	75,000	5°	5°	100	MS18005-4435
4530	4530	26.0	135	100,000	11°	11°	50	MS18005-4530
4551	4551	28.0	250	65,000	6°	2°	25	MS24517-4551
4553	4553	28.0	250	225,000	2°	2°	25	MS25241-4553

* AT DESIGN VOLTS

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3.5.3 Support arm. A support arm with a mounting flange suitable for mounting the spotlight on a plane surface shall be provided. The support arm shall support and position the lamp head under all conditions specified herein. The length of the arm shall be such that the distance from the horizontal center line of the lamp housing to the face of the mounting flange shall be 6 1/2 inches plus or minus 1/4 inch. Mounting screws or bolts of sufficient length to fasten the arm and adapter (see 3.5.6) to a panel or bracket 1/8 inch in thickness shall be furnished complete with self-locking nuts or nuts and lockwashers.

3.5.4 Control arm and handle. A control arm and handle as specified in 1.2 and 6.2 shall be furnished. The arm shall be an integral part of or fastened to the support arm and flange (3.5.3). The control arm, when coupled to the support arm shall completely enclose all moving parts. The length of the control arm given in paragraph 1.2 classes A and B shall be plus or minus 1/8" measured from the mounting surface of the mounting flange (see 3.5.3) to the horizontal center line of the control handle or knob. Control knobs shall not exceed 4" diameter. Control handles shall not extend more than 5" from the vertical center line of the control arm. The operations of the control knob or handle shall give the movement specified in 3.3.1.

*3.5.5 Switch and wiring. The switch and electrical conductors shall be furnished to meet electrical voltage and current requirements specified for particular size from Table I. The switch shall be incorporated into the control handle or bracket mounted on the control arm. Brackets, when furnished, shall be positionable at least 270 degrees about the arm. Provisions shall be made to permit electrical power connection to the spotlight after the installation is completed. Wires shall be of a length to permit bulb installation and connection with the bulb rotated 90° from its installed position.

*3.5.6 Roof seal. A sealing gasket to fill the voids remaining when the mounting flange contacts a convex surface as found on commercial vehicle roofs shall be furnished.

3.6 Lubrication. All internal moving parts shall be lubricated prior to assembly. All openings shall be sealed to prevent leakage. The light shall not require subsequent lubrication.

*3.7 Finishes and protective coatings. All finishes and protective coatings shall be in conformance to MIL-STD-808.

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3.7.1 Type II. Type II lights shall be prime painted on the exposed surfaces of exterior assemblies, lamphead, support arm, etc. Other treatment shall be as Type I.

3.7.2 Type I. Type I spotlights and interior parts of Type II spotlights shall be furnished as follows:

a. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except small holes need not be anodized.

b. Aluminum alloy which are not subject to abrasion and which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. Ferrous parts, except those fabricated from corrosion-resistant steel, shall be cadmium plated in accordance with QQ-P-416.

d. All parts normally painted or plated in commercial practice shall be treated in accordance with the manufacturer's standard practice if not in conflict with 3.7.1 or the balance of this paragraph.

3.8 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.8.1 Nameplate. A nameplate shall be permanently fastened to the lamp housing approximately 180 degrees from the mounting containing the following information:

Spotlight, Roof-Mounted
Max. volts
Max. Amps
Manufacturer's Model Number
Manufacturer's Name
US
* Military Part Number

3.9 Instructions. Mounting instructions, including required templates shall be packed with each light.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless otherwise disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspections, consisting of examinations and tests, shall be classified as follows:

a. Preproduction inspection(see 4.3).

b. Acceptance inspection (see 4.4).

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4.3 Preproduction inspection. A sample of the light the supplier proposes to furnish under this specification shall be examined and tested as follows:

4.3.1 The light shall be examined to assure conformance to all requirements of section 3 of this specification.

4.3.2 Operation. The light shall be operated thru the full angular range specified in 3.3.1 with the lamp on. Conformance with bulb position in 3.5.1 shall be verified. No binding or looseness of parts shall be evident. The lamp head shall retain its position when the control handle is released.

4.3.3 Environmental inspection. Unless otherwise specified (see 6.2), the light with lamp installed shall be subjected to the following tests:

4.3.3.1 Failed lamps shall not constitute light failure. The lamp shall be replaced and the tests continued.

4.3.3.2 High temperature. The light shall be placed in a test chamber having a temperature of 71 degrees centigrade plus or minus 1 degree, with a relative humidity of not more than 15 percent. After a soaking period of not less than 24 hours the light shall be operated in accordance with paragraph 4.3.2. The light shall then be allowed to return to normal ambient conditions and shall be examined in accordance with 4.3.1 and operated in accordance with 4.3.2.

4.3.3.3 Low temperature. The light shall be placed in a test chamber having a temperature of minus 54° centigrade. After a soaking period of not less than 24 hours the spotlight shall be operated in accordance with 4.3.2. Twice, first with the lamp off, then with the lamp on. The light shall be allowed to return to normal ambient conditions and examined in accordance with 4.3.2.

4.3.3.4 Vibration. The spotlight shall be subjected to vibration in accordance with vibration test of SAE J575. Upon completion of the test the light shall be examined in accordance with 4.3.1 and operated in accordance with 4.3.2.

4.3.3.5 Moisture. The spotlight shall be subjected to moisture in accordance with moisture test of SAE J575. Upon completion of this test light shall be operated in accordance with 4.3.2.

4.3.3.6. Dust. The spotlight shall be subjected to dust in accordance with dust test of SAE J575. Upon completion of this test, light shall be operated in accordance with 4.3.2.

4.3.3.7. Corrosion. The spotlight shall be subjected to corrosion in accordance with corrosion test of SAE J575. Upon completion of this test the spotlight shall be operated in accordance with 4.3.2.

4.3.4 Ease of maintenance. Such components as the sealed beam unit, control handle, switch, gears, adapter, and wiring shall be removed and reinstalled and those adjustments necessary shall be made to determine ease of maintenance without special tools.

4.3.5 Examination of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to insure compliance with this specification section 5 and the contract.

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4.4 Acceptance tests. Each spotlight shall be subjected to the tests in 4.3.1 and 4.3.2 before acceptance.

*5 PACKAGING.

5.1 General. Preservation, packaging, and packing pertains only to direct purchases by the Government. When the spotlight is furnished as part of a vehicle or other end item, these provisions do not apply (see 6.2).

5.2 Preservation and packaging.

5.2.1 Level A. Each spotlight shall be preserved in acordance with MIL-P-116, Method II, and individually packaged in containers conforming to PPP-B-636. The container shall be sealed at all joints and seams, including the manufacturers joint, with tape conforming to PPP-T-60.

5.2.2 Level C. Spotlights shall be preserved and packaged in accordance with the manufacturers commercial paractice.

5.3 Intermediate packaging. Only identical items shall be included in an intermediate package. For items preserved and packaged in accordance with 5.2.1. The requirements for intermediate packaging shall be at the option of the contractor. Spotlights preserved and packaged in accordance with 5.2.2 shall be packaged in containers conforming to PPP-B-566, PPP-B-636, PPP-B-676, or PPP-B-665. Unless otherwise specified by the procuring activity (see 6.2) the quantity of unit packages to be included in each intermediate package shall be at the option of the contractor and as governed by the limitations of the container being used.

5.4 Packing.

*5.4.1 Level A. Spotlights preserved and packaged as specified in 5.2.1 and 5.3 shall be packed in export exterior-type shipping containers conforming to PPP-B-585, PPP-B-601, PPP-B-621, or PPP-B-636. The gross weight of wood or wooden type containers shall not exceed 210 pounds. Fiberboard containers shall not exceed the weight and size limitations of the box specification. Exterior containers shall be of uniform size and shape, of minimum tare and cube consistent with the protection required, and shall contain identical quantities. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

*5.4.2 Level B. Spotlights, preserved and packaged as specified in 5.2.1 and 5.3 shall be packed in domestic-type exterior containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621 or weather resistant fiberboard in accordance with PPP-B-636. When fiberboard containers are used, the fiberboard shall conform to the special requirements table of PPP-B-636, as applicable. Exterior containers shall be of minimum cube and tare consistent with the protection required. Exterior containers shall be of uniform size and shape and shall contain identical quantities. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. The gross weight of each pack shall not exceed 510 pounds except fiberboard cartons, the gross weight shall not exceed the weight limitations of the box specification.

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5.4.3 Level C. Packages which require over packing shall be packed to insure carrier acceptance and safe delivery to destination in compliance with Official Classification Committee, Uniform Freight Classification Rules, or American Trucking Association, National Freight Classification Rules or with other carrier rules and regulations applicable to the mode of transportation.

5.5 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. Class A spotlights are intended for universal use including driver operated mountings. Class B spotlights are intended for use from stationary vehicles and are not to be cab roof mounted.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type, class and size required (see 1.2).

c. Environmental inspections, if not required (see 4.3.3).

d. Level of preservation, packaging and packing required (see sec. 5).

*6.3 Part numbering. The old and new part number system is cross referenced in Table II. The new part number shall consist of the letter M followed by 4510, and a dash number. The dash number will consist of 7 digits. The first digit designates type, the second class, the third through sixth, size. The sizes covered by this specification are listed in Table I.

Example: MIL-S-4510, Type I, Class B,
Size 4435
Part Number: M4510-B4435

*6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - ME
Navy - YD
Air Force - 99

User Activity:

Navy - MC

Preparing Activity

Air Force - 82

Project 6220-0306

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TABLE II. Cross reference of MIL-S-4510D to MIL-S-4510E part numbers.

MIL-S-4510D	MIL-S-4510E
MIL-S-4510 Type I Class A Size 1	M4510-1A4535
MIL-S-4510 Type I Class B Size 1	M4510-1B4535
MIL-S-4510 Type II Class A Size 1	M4510-2A4535
MIL-S-4510 Type II Class B Size 1	M4510-2B4535
MIL-S-4510 Type I Class B Size 2	M4510-1B4435
MIL-S-4510 Type II Class B Size 2	M4510-2A4435
MIL-S-4510 Type II Class B Size 2	M4510-2B4435
MIL-S-4510 Type I Class A Size 3	M4510-1A4530
MIL-S-4510 Type I Class B Size 3	M4510-1B4530
MIL-S-4510 Type II Class A Size 3	M4510-2A4530
MIL-S-4510 Type II Class B Size 3	M4510-2B4530

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DOCUMENT IDENTIFIER AND TITLE

MIL S 4570 SPOTLIGHT, INCANDESCENT, ROOF MOUNTED, AUTOMOTIVE.

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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